# STAGING AREA DESIGN

## Olist E-Commerce Business Intelligence Project

**Version:** 1.0
**Date:** August 2026
**Database:** PostgreSQL
**Schema:** `staging`
**Source:** Olist Brazilian E-Commerce Dataset

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# 1. Introduction

## 1.1 Purpose

The Staging Area is an intermediate storage layer used to receive and temporarily store data extracted from the source systems before its integration into the Data Warehouse.

For this project, the Staging Area is implemented in PostgreSQL using a dedicated `staging` schema.

The data stored in this layer remains as close as possible to the original source structure.

The Staging Area does not contain business-oriented transformations or analytical calculations.

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## 1.2 Role in the BI Architecture

The Staging Area is positioned between the source data and the Data Warehouse.

```
Olist Dataset
      │
      ▼
CSV Files
      │
      ▼
PostgreSQL
   staging
      │
      ▼
Data Warehouse
      │
      ├──► Power BI
      ├──► Machine Learning
      └──► n8n Automation
```

The Staging Area therefore provides a controlled intermediate layer between the source data and the analytical environment.

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# 2. Objectives

The main objectives of the Staging Area are:

- Centralize the source data in PostgreSQL.
- Provide an intermediate storage layer before the Data Warehouse.
- Preserve the original source information.
- Maintain a structure close to the source datasets.
- Separate raw source data from business-oriented data models.
- Facilitate data validation and quality checking.
- Provide a stable source for the Data Warehouse loading process.

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# 3. Source Data

The project uses the Olist Brazilian E-Commerce Dataset.

The dataset contains several CSV files representing different business entities and events.

The main source files are:

| Source File | Description |
|---|---|
| `olist_customers_dataset.csv` | Customer information |
| `olist_orders_dataset.csv` | Order information |
| `olist_order_items_dataset.csv` | Items associated with orders |
| `olist_products_dataset.csv` | Product information |
| `olist_sellers_dataset.csv` | Seller information |
| `olist_order_payments_dataset.csv` | Payment information |
| `olist_order_reviews_dataset.csv` | Customer reviews |
| `olist_geolocation_dataset.csv` | Geographic information |
| `product_category_name_translation.csv` | Product category translations |

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# 4. PostgreSQL Staging Environment

The Staging Area is implemented in PostgreSQL.

A dedicated schema named `staging` is used to isolate the source-oriented tables from the future Data Warehouse structures.

The logical organization is:

```
PostgreSQL Database
│
├── staging
│   ├── customers
│   ├── orders
│   ├── order_items
│   ├── products
│   ├── sellers
│   ├── order_payments
│   ├── order_reviews
│   ├── geolocation
│   └── category_translation
│
└── dw
    ├── dimensions
    └── facts
```

The `dw` schema represents the future Data Warehouse layer.

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# 5. Staging Tables

The following tables are stored in the `staging` schema.

| Table | Source | Description |
| --- | --- | --- |
| `customers` | Olist Customers Dataset | Customer information |
| `orders` | Olist Orders Dataset | Order information |
| `order_items` | Olist Order Items Dataset | Items purchased within orders |
| `products` | Olist Products Dataset | Product information |
| `sellers` | Olist Sellers Dataset | Seller information |
| `order_payments` | Olist Order Payments Dataset | Payment information |
| `order_reviews` | Olist Order Reviews Dataset | Customer review information |
| `geolocation` | Olist Geolocation Dataset | Geographic information |
| `category_translation` | Category Translation Dataset | Product category translations |

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# 6. Data Characteristics

The Staging Area follows a source-oriented data model.

The data is stored with minimal structural modification.

The following principles are applied:

### 6.1 Source Preservation

Source records are preserved as closely as possible in the staging tables.

### 6.2 No Business Rules

Business rules are not applied in the Staging Area.

For example, business metrics such as:

- Total Revenue
- Average Order Value
- Customer Lifetime Value
- Delivery Performance

are not calculated in the staging layer.

These calculations belong to the analytical/Data Warehouse layer.

### 6.3 No Dimensional Modeling

The Staging Area does not use a Star Schema.

There are no:

- `fact_*` tables
- `dim_*` tables
- Surrogate keys created for analytical purposes

The dimensional model will be implemented in the Data Warehouse.

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# 7. Data Transformation Policy

The Staging Area follows a **minimal-transformation principle**.

The objective is to keep the data close to the source.

No business transformation is performed at this stage.

The following types of transformations are intentionally excluded:

- Business calculations
- Aggregations
- KPI calculations
- Business classification
- Fact/dimension creation
- Surrogate key generation
- Analytical joins
- Business-level filtering

The Staging Area therefore acts primarily as a **raw/intermediate storage layer**.

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# 8. Data Quality

Although the data is not transformed at the staging level, basic technical validation is important.

The following checks can be performed:

### 8.1 Record Count

The number of records in the source file can be compared with the number of records stored in the corresponding staging table.

### 8.2 Structure Validation

The structure of the staging table should correspond to the source dataset.

Checks include:

- Number of columns
- Column names
- Data types
- Expected fields

### 8.3 Null Value Identification

Important identifier fields should be checked for unexpected missing values.

Examples:

```
customer_id
order_id
product_id
seller_id
```

### 8.4 Duplicate Identification

Potential duplicate records should be identified according to the characteristics of each source dataset.

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# 9. Naming Convention

The Staging Area follows a consistent naming convention.

## Schema

```
staging
```

## Tables

Tables use lowercase `snake_case`.

Examples:

```
customers
orders
order_items
order_payments
order_reviews
products
sellers
```

## Columns

Columns also use lowercase `snake_case`.

Examples:

```
customer_id
order_id
product_id
seller_id
order_purchase_timestamp
```

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# 10. Data Dictionary

A separate Data Dictionary is maintained to describe the structure of the staging tables.

The Data Dictionary contains information such as:

- Table name
- Column name
- Data type
- Nullable status
- Column description
- Source column
- Key information

The file is maintained separately as:

```
Data_Dictionary.xlsx
```

This provides a detailed technical reference for the staging layer.

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# 11. Staging to Data Warehouse

The Staging Area serves as the source layer for the future Data Warehouse.

The transition can be represented as:

```
STAGING
│
├── customers
├── orders
├── order_items
├── products
├── sellers
├── order_payments
├── order_reviews
├── geolocation
└── category_translation
        │
        ▼
Business Transformations
        │
        ▼
DATA WAREHOUSE
│
├── dim_customer
├── dim_product
├── dim_seller
├── dim_date
└── fact_sales
```

Business transformations, data integration and dimensional modeling are therefore performed during the Staging-to-Data-Warehouse phase.

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# 12. Governance and Access

The Staging Area is considered a technical layer of the BI architecture.

Access should primarily be provided to:

- Data Engineers
- BI Developers
- ETL processes
- Technical users responsible for data validation

Business users should generally consume data from the Data Warehouse and analytical layers rather than directly from staging tables.

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# 13. Benefits of the Staging Area

The implementation of a dedicated Staging Area provides several benefits:

### Separation

Source data is separated from analytical data.

### Traceability

The Data Warehouse can be traced back to the source-oriented staging layer.

### Reliability

The Data Warehouse does not depend directly on CSV files.

### Maintainability

Changes to the source ingestion process can be isolated from the analytical layer.

### Data Quality

Technical data validation can be performed before data is integrated into the Data Warehouse.

### Scalability

The architecture can later accommodate additional source systems without directly coupling them to the reporting layer.

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# 14. Final Architecture

The current architecture of the project can be summarized as follows:

```
┌──────────────────────────────┐
│      Olist Dataset           │
│          CSV Files           │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│       Data Ingestion         │
│           Talend             │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│       PostgreSQL             │
│          STAGING               │
│                              │
│  customers                   │
│  orders                      │
│  order_items                 │
│  products                    │
│  sellers                     │
│  order_payments              │
│  order_reviews               │
│  geolocation                 │
│  category_translation        │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│      Data Warehouse          │
│       Star Schema            │
└──────────────┬───────────────┘
               │
        ┌──────┼───────┐
        ▼      ▼       ▼
    Power BI   ML      n8n
```

# 15. Conclusion

The Staging Area provides a dedicated technical layer for storing the Olist source data in PostgreSQL before its integration into the Data Warehouse.

The staging tables remain close to the original source structure and do not contain business-oriented transformations.

This approach provides a clear separation between:

**Source Data → Staging → Data Warehouse → Analytics**

The next phase of the project is the **Data Warehouse Design**, including the identification of facts and dimensions, the Star Schema design and the definition of the Data Warehouse tables.